Zephyr Real time OS

1. Links

- Zephyr Project Home page

2. Installation

2.1. Zephyr Installation

Followed the instructions at: Getting started

```
winget install Kitware.CMake Ninja-build.Ninja oss-winget.gperf python Git.Git oss-winget.dtc wget 7zip.7zip

Added 7Zip to the path

cd D:\skunk_zephyr
python -m venv .venv

D:\skunk_zephyr\.venv\Scripts\activate.bat

pip install west

west init zephyrproject
cd zephyrproject
west update

Takes a while

west zephyr-export

west packages pip --install

west sdk install --toolchains arm-zephyr-eabi

cd zephyr
west build -p always -b nucleo_h723zg samples\basic\blinky

set PATH=%PATH%;D:\STM32Cube\STM32CubeProgrammer\bin

west flash
```

2.2. How to update the Zephyr Installation

```
D:\skunk_zephyr\.venv\Scripts\activate.bat

cd zephyrproject
west update
```

2.3. The Samples / Blinky

The Board is Nucleo H723ZG: Zephyr support for Nucleo H723ZG

- Board name: nucleo_h723zg
- Architecture: arm
- SoC: stm32h723xx

To debug with PuTTY, the prj.conf file is set as:

```
CONFIG_GPIO=y
CONFIG_UART_CONSOLE=y
CONFIG_CONSOLE=y
CONFIG_PRINTK=y
CONFIG_STDOUT_CONSOLE=y
```

A Workspace application. This creates a new workspace.

```
CD D:\skunk_zephyr\zephyrproject
git clone https://github.com/zephyrproject-rtos/example-application my-app
```

3. Application Development

The debugger is at:

```
C:\Users\diego\zephyr-sdk-0.17.4\arm-zephyr-eabi\bin\arm-zephyr-eabi-gdb.exe
```

Output file is at:

```
D:/skunk_zephyr/zephyrproject/build/zephyr/zephyr.elf
```

The command to debug an application has the format:
path/to/gdb path/to/zephyr.elf
So:

```
C:\Users\<USER>\zephyr-sdk-0.17.4\arm-zephyr-eabi\bin\arm-zephyr-eabi-gdb.exe <ZEPHYRPROJECT>/build/zephyr/zephyr.elf
```